Pre-Analysis Plan for the #ManyDaughters Project

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Overview

This document outlines our approach to testing four hypotheses concerning the relationship between having daughters and gender-related attitudes and behaviors, using SOEP-Core data (2000–2020).

Hypotheses

H1. Having daughters increases support for gender equality (attitudes).

H2. Having daughters increases left-leaning political preferences.

H3. Having daughters increases gender equality in household behaviors (chores, childcare, parental leave).

H4. The effect of having daughters on gender attitudes is stronger for men than for women.

Data and Sample

- **Dataset**: SOEP-Core, waves 2000–2020.
- **Sample**: Respondents aged 18 and older, with at least one biological child and valid values for treatment and outcome variables.

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- **Treatment Variable**: Binary indicator "`has_daughter`" equals 1 if the respondent has at least one biological daughter in the household, 0 otherwise. Robustness check: restrict to one-child households.

Control Variables (Used in All Models)

The control variables and variable names in SOEP data are as follows: Age (constructed by substracting birth year variable birth_v2 from 2025), education (plg0268), household income (hlc0005_h), employment status (plb0022_h), marital status (pld0131_v2), migration background (first foreign citizenship: plj0014_v1), year fixed effects. Gender (pla0009_h) included where relevant.

H1: Gender Equality Attitudes

Variables:

- **Outcome: Gender-equality attitudes**
  We interpret gender equality as equality for all genders and sexual preferences, in line with the German Federal Ministry's understanding of gender equality as encompassing the full participation and recognition of people regardless of gender identity or sexual orientation (Federal Ministry for Family Affairs, Senior Citizens, Women and Youth (n.d.)). Therefore, we construct a scale using six SOEP variables from the 2018[1] wave (on a 7-point Likert scale, where 1 = "Do not agree at all" and 7 = "Completely agree"):
  - `plh0357` – Legal recognition of same-sex marriage
  - `plh0298_v2` – Children under 6 suffer if their mother works *(reverse-coded)*
  - `plh0302_v2` – Children under 3 suffer if their mother works *(reverse-coded)*
  - `plh0308_v2` – Equal work-sharing between men and women
  - `plh0359` – Same-sex couples can raise children just as well
  - `plh0360` – Transgender people should be recognized as normal

[1]The gender attitude questions used to construct our outcome variable were only included in the 2018 wave of the SOEP. As a result, our analysis for Hypothesis 1 is necessarily limited to respondents who completed the individual questionnaire in 2018. Although the broader project sample spans the years 2000 to 2020, the attitude data required for this hypothesis are unavailable in other waves. Therefore, our analysis is cross-sectional, using data from a single year (2018).

We compute the respondent's mean score (after reverse-coding the relevant items). The resulting continuous variable, `attitude_mean_i`, ranges from 1 to 7. We retain observations where at least five of the six items are non-missing.

- **Key independent variable:** `has_daughter_i` is a binary indicator equal to 1 if the respondent has at least one biological daughter living in the household in 2018; 0 otherwise.

Internal Consistency Check: Before estimating the main model, we will compute Cronbach's alpha for the six-item scale (after reverse-coding). The procedure is:

1. Calculate Cronbach's α on the 2018 sample.
2. If $\alpha \geq 0.70$, retain all six items.
3. If $\alpha < 0.70$, iteratively drop the item that improves α the most, continuing until $\alpha \geq 0.70$ or at least three items remain.[2]
4. Document any item removals and use the final composite in all main analyses.

Sample and Exclusion Criteria:

- Sample is restricted to individuals who completed the SOEP individual questionnaire in 2018.
- Respondents must have answered at least five of the six attitude items.
- Missing values in control variables are handled via mean imputation (for continuous variables) and missing-value dummies (for categorical variables).

Exact Analysis Specification:
We estimate the following cross-sectional regression model:

$$\texttt{attitude_mean}_i = \beta_0 + \beta_1 \, \texttt{has_daughter}_i + \mathbf{X}_i^\top \boldsymbol{\gamma} + \varepsilon_i,$$

where:

- $\texttt{attitude_mean}_i$ is the composite attitude score.
- $\texttt{has_daughter}_i$ is the main independent variable.

[2] In this case, we will follow Stöcker et al. (2024) and include only the following three items to construct the outcome variable: `plh0298_v2`, `plh0302_v2`, and `plh0308_v2`.

- $\mathbf{X}_i$ is the vector of control variables.
- Robust (heteroskedasticity-consistent) standard errors are used.

Interpretation and Inference:

- Primary statistic: the coefficient $\hat{\beta}_1$ on `has_daughter`.
- Hypothesis support: If $\hat{\beta}_1 > 0$ and $p < 0.05$, we interpret this as evidence in favor of the hypothesis.
- Confidence intervals and exact p-values will be reported.

Deviations: If deviations are necessary, they will be documented in a separate appendix and shared with the project coordinators.

H2: Left-Leaning Political Preferences

Outcome: Reversed political orientation scale (e.g., `plh0004`) such that higher values reflect more left-leaning views.
Model:

$$Y_{it} = \beta_0 + \beta_1 \texttt{has_daughter}_{it} + \beta_2 \texttt{controls}_{it} + \delta_t + \epsilon_{it}$$

Hypothesis: $\beta_1 > 0$

H3: Gender Equality in Household Behaviors

Outcomes:

We focus on the following categories of household behavior, reported separately for weekdays, Saturdays, and Sundays, and distinguished by mothers and fathers:

- Time spent on childcare (`pli0044_h`, `pli0019_h`, `pli0022_h`)
- Time spent on household chores (`pli0012_h`, `pli0043_h`, `pli0016_h`)
- Time spent on repair work (`pli0031_h`, `pli0034_v1`, `pli0049_h`)

Each activity is reported using three separate variables: average time spent on a weekday, Saturday, and Sunday. We convert these into total weekly hours using the following formula:

$$\text{Weekly hours} = 5 \times \text{weekday average} + 1 \times \text{Saturday} + 1 \times \text{Sunday}$$

We compute weekly totals separately for mothers and fathers, for each activity.

Construction of Outcome Variable:

For each household i in year t, we compute the total weekly hours spent on all the above activities by the mother (M_{it}^{mom}) and by the father (M_{it}^{dad}). The outcome variable is then defined as:

$$Y_{it} = M_{it}^{\text{mom}} - M_{it}^{\text{dad}}$$

We exclude all household-year observations where (i) the parents are divorced, or (ii) any time use data is missing for either parent.

Model Specification:

$$Y_{it} = \beta_0 + \beta_1 \texttt{has_daughter}_{it} + \mathbf{X}_{it}^{\text{mom}}\boldsymbol{\beta}_2 + \mathbf{X}_{it}^{\text{dad}}\boldsymbol{\beta}_3 + \mathbf{X}_{it}^{\text{hh}}\boldsymbol{\beta}_4 + \delta_t + \epsilon_{it}$$

where:

- $\texttt{has_daughter}_{it}$ is an indicator for whether the household has at least one daughter.
- $\mathbf{X}_{it}^{\text{mom}}$ and $\mathbf{X}_{it}^{\text{dad}}$ are vectors of parent-specific controls (e.g., age, education, employment status).
- $\mathbf{X}_{it}^{\text{hh}}$ includes household-level controls (e.g., household income).
- δ_t are year fixed effects.

Standard errors are clustered at the household level.

Hypothesis: $\beta_1 > 0$ (Having a daughter is associated with a more equal division of unpaid household labor.)

Robustness: Household Fixed Effects

We will estimate a version of the model with household fixed effects:

$$Y_{it} = \beta_1 \texttt{has_daughter}_{it} + \mathbf{X}_{it}^{\text{mom}}\boldsymbol{\beta}_2 + \mathbf{X}_{it}^{\text{dad}}\boldsymbol{\beta}_3 + \mathbf{X}_{it}^{\text{hh}}\boldsymbol{\beta}_4 + \delta_t + \mu_i + \epsilon_{it}$$

where μ_i denotes household fixed effects.

Because the indicator `has_daughter` is time-invariant for many households, this model will be estimated on the subset of households whose child composition changes during the panel (e.g., those who have their first child while observed). This allows us to identify

the effect of the birth of a daughter on the gender division of household labor, net of all time-invariant household characteristics.

Standard errors will be clustered at the household level.

H4: Gender-Differentiated Attitude Effects

Variables:

- **Outcome: Gender-equality attitudes**
  As in Hypothesis 1, we construct a composite scale using six SOEP variables from the 2018 wave:
  - `plh0357`, `plh0298_v2`, `plh0302_v2`, `plh0308_v2`, `plh0359`, `plh0360`

  Items `plh0298_v2` and `plh0302_v2` are reverse-coded. The composite score `attitude_mean_i` is the respondent's mean across all available items (minimum of five non-missing values required). The final score ranges from 1 to 7, with higher values indicating greater support for gender equality.

- **Key independent variable:** `has_daughter_i` is a binary indicator equal to 1 if the respondent has at least one biological daughter living in the household in 2018; 0 otherwise.

- **Moderator:** `male_i` is a binary indicator equal to 1 for male respondents and 0 for female respondents. For this variable in the model, we will use the gender variable (pla0009_h) mentioned in control variables section.

Internal Consistency Check:

Before estimating the main model, we will compute Cronbach's alpha for the six-item scale (after reverse-coding). The procedure is:

1. Calculate Cronbach's α on the 2018 sample.
2. If $\alpha \geq 0.70$, retain all six items.
3. If $\alpha < 0.70$, iteratively drop the item that improves α the most, continuing until $\alpha \geq 0.70$ or at least three items remain.[3]
4. Use the resulting scale to construct the final `attitude_mean_i` variable.

[3]We will apply the same procedure with Hypothesis 1. See explanation in Footnote 1.

Sample and Exclusion Criteria:

- The sample is limited to respondents who completed the SOEP individual questionnaire in 2018.
- Observations are retained if the respondent has non-missing values for gender, daughter status, and at least five out of the six attitude items.

Exact Analysis Specification:
We estimate the following cross-sectional OLS regression model:

$$\texttt{attitude_mean}_i = \beta_0 + \beta_1 \texttt{has_daughter}_i + \beta_2 \texttt{male}_i + \beta_3 (\texttt{has_daughter}_i \times \texttt{male}_i) + \varepsilon_i$$

where:

- $\texttt{attitude_mean}_i$ is the composite gender attitude score.
- $\texttt{has_daughter}_i$ is the main independent variable.
- $\texttt{male}_i$ indicates respondent gender.
- The coefficient β_3 captures the differential effect of having daughters for men compared to women.
- Robust (heteroskedasticity-consistent) standard errors will be used.

Interpretation and Inference:

- The key test statistic is the interaction term $\hat{\beta}_3$.
- Hypothesis support: If $\hat{\beta}_3 > 0$ and $p < 0.05$, we interpret this as evidence supporting H4.
- Confidence intervals and exact p-values will be reported.

Deviations: Any required deviations will be documented in a separate appendix and reported to the project coordinators.

Estimation and Reporting

- Each hypothesis will be tested via a separate regression model.
- We will include regression coefficient, SE, test statistic, p-value, N, and number of individuals.

References

Federal Ministry for Family Affairs, Senior Citizens, Women and Youth (n.d.). *Gender equality policy.* https://www.bmfsfj.de/bmfsfj/meta/en/equality/gender-equality-policy.

Stöcker, A.-K., Gauglitz, I. K., and Schütz, A. (2024). Managers' traditional gender role attitudes. *Journal of Personnel Psychology.*